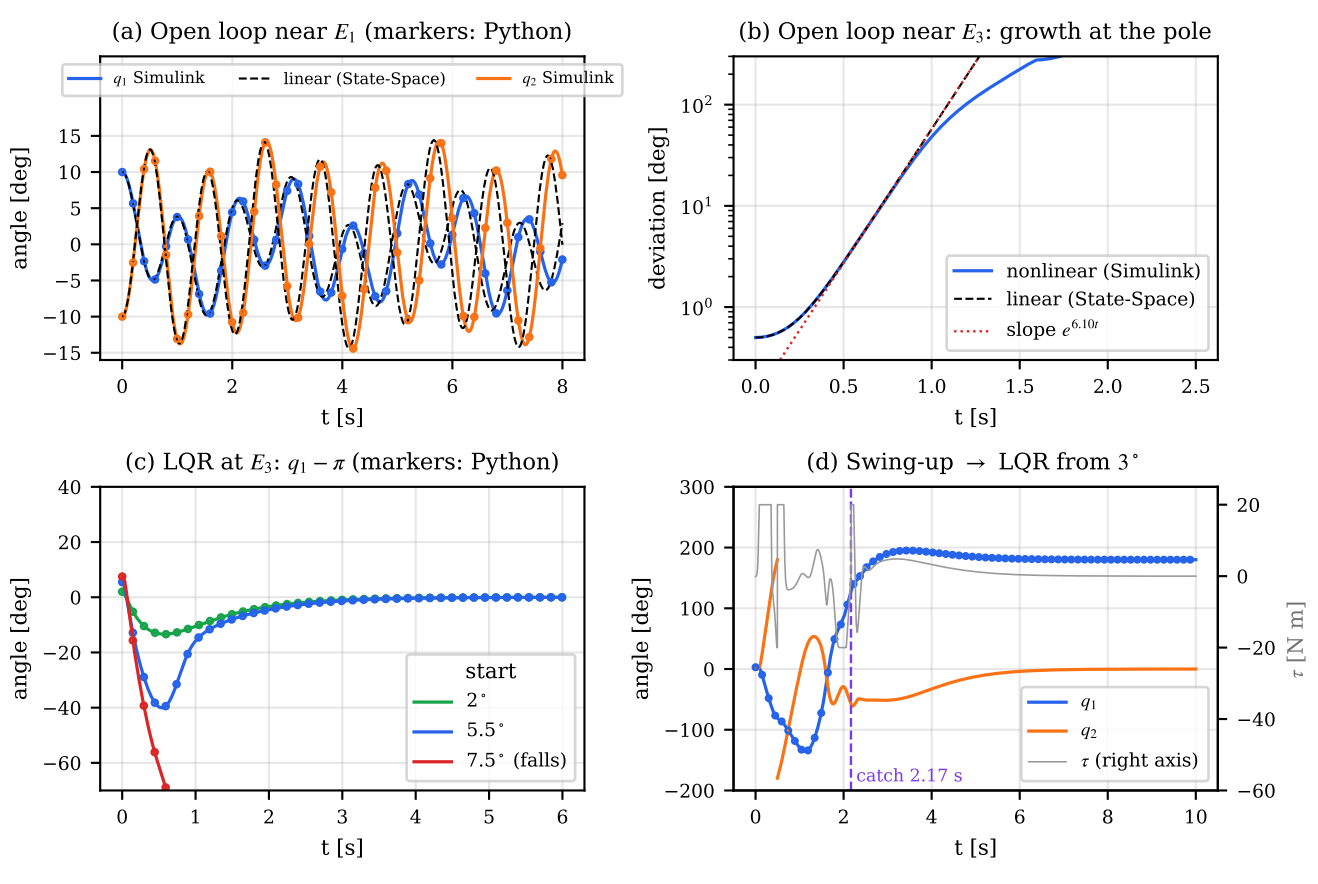
(a) Open loop near E_1 (markers: Python)
q_1 Simulink
linear (State-Space)
q_2 Simulink
angle [deg]
t [s]
(b) Open loop near E_3: growth at the pole
deviation [deg]
nonlinear (Simulink)
linear (State-Space)
slope $e^{6.10t}$
t [s]
(c) LQR at E_3: $q_1 - \pi$ (markers: Python)
angle [deg]
start
2°
5.5°
7.5° (falls)
t [s]
(d) Swing-up → LQR from 3°
angle [deg]
τ [N m]
q_1
q_2
τ (right axis)
catch 2.17 s
t [s]